

A Wolfram-Language Derivation of the Coupling Chain

Proposition 5.3 of *The Electromagnetic Sphere*

Step-by-step machine verification in $\text{Cl}(3, 3) \cong M_8(\mathbb{R})$

Verification note for the ES monograph (Part I)

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Abstract

This note reconstructs Proposition 5.3 (the *coupling chain*) of the Electromagnetic Sphere monograph as an explicit, reproducible sequence of Wolfram Language computations. Every numerical and algebraic claim is anchored to the literal output of a stateless Wolfram kernel acting on the faithful real 8×8 representation of $\text{Cl}(3, 3)$. The chain establishes that the photon generator J is forced uniquely as the centralizer of the kernel clock $\hat{W}$ inside $\mathfrak{su}(2)_T$, that the Hermitian charge operator $Q = -iJ$ has spectrum $\{(\pm 1)^4\}$, that identifying the real 8-spinor with a particle–antiparticle pair quantizes the elementary charge as $q = \pm \frac{1}{2}$ (in units of g), and hence that $e_0 = g/2$ and $\alpha_{\text{em}} = g^2/16\pi$, with the single gauge coupling g the only free parameter. Status of the chain: **[EXACT]** for the spectral and algebraic facts, **[DERIVED]** for the physical charge assignment, **[OPEN]** for the magnitude of g .

1 Setup: the algebra and its real representation

The framework is built on six orthonormal generators $e_1, \dots, e_6$ obeying the Clifford relation

$$\{e_a, e_b\} = 2 \eta_{ab} \mathbf{1}, \quad \eta = \text{diag}(+1, +1, +1, -1, -1, -1),$$

with e_1, e_2, e_3 *temporal* ($e_i^2 = +\mathbf{1}$) and e_4, e_5, e_6 *spatial* ($e_i^2 = -\mathbf{1}$). They generate the real Clifford algebra $\text{Cl}(3, 3)$, of dimension $2^6 = 64$. A faithful real representation is the Jordan–Wigner assignment

$$\begin{aligned} e_1 &= \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, & e_2 &= \sigma_z \otimes \sigma_x \otimes \mathbf{1}, & e_3 &= \sigma_z \otimes \sigma_z \otimes \sigma_x, \\ e_4 &= \varepsilon \otimes \mathbf{1} \otimes \mathbf{1}, & e_5 &= \sigma_z \otimes \varepsilon \otimes \mathbf{1}, & e_6 &= \sigma_z \otimes \sigma_z \otimes \varepsilon, \end{aligned}$$

with the real Pauli matrices $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\varepsilon = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Verification of the representation. We build the six generators by Kronecker product, confirm the Clifford relation and the signature of the squares, and check that the 64 ordered blades are linearly independent (rank 64) in $M_8(\mathbb{R}) \cong \mathbb{R}^{64}$, i.e. $\text{Cl}(3, 3) \cong M_8(\mathbb{R})$.

Wolfram Language input

```

sx = {{0,1},{1,0}}; sz = {{1,0},{0,-1}}; ep = {{0,-1},{1,0}}; id = IdentityMatrix[2];
ee = {KroneckerProduct[sx,id,id], KroneckerProduct[sz,sx,id], KroneckerProduct[sz,sz,sx],
      KroneckerProduct[ep,id,id], KroneckerProduct[sz,ep,id], KroneckerProduct[sz,sz,ep]};
g[a_] := ee[[a]]; I8 = IdentityMatrix[8];
eta = DiagonalMatrix[{1,1,1,-1,-1,-1}];
```

```
(* Clifford relation {e_a,e_b}=2 eta_ab I8 *)
cliffOK = AllTrue[Flatten@Table[(g[a].g[b]+g[b].g[a]) - 2 eta[[a,b]] I8, {a,6},{b,6}], # == 0
&];
(* signature of squares *)
squares = Table[Tr[g[a].g[a]]/8, {a,6}];
(* rank of the 64 ordered blades in R^64 *)
prod[lst_] := If[lst === {}, I8, Dot @@ (g /@ lst)];
bladeMats = prod /@ Subsets[Range[6]];
{"Clifford 2eta" -> cliffOK, "squares" -> squares,
"#blades" -> Length[bladeMats], "rank" -> MatrixRank[Flatten /@ bladeMats]}
```

Literal Wolfram output

```
{"Clifford 2eta" -> True, "squares" -> {1, 1, 1, -1, -1, -1},
"#blades" -> 64, "rank in R^64" -> 64, "Cl(3,3)=M8(R)" -> True}
```

The kernel, the temporal-rotation algebra, and the photon are

$$\hat{W} = \frac{1}{\sqrt{3}}(e_1 + e_2 + e_3), \quad L_1 = e_2 e_3, \quad L_2 = e_3 e_1, \quad L_3 = e_1 e_2, \quad J = \frac{1}{\sqrt{3}}(L_1 + L_2 + L_3).$$

2 The coupling chain (Proposition 5.3)

Proposition 1 (Coupling chain, monograph 5.3). *In the real 8-spinor the photon $U(1)_J$ is generated by J , the unique unit element of $\mathfrak{su}(2)_T$ commuting with the kernel $\hat{W}$. The associated Hermitian charge $Q = -iJ$ has eigenvalues ± 1 , each fourfold; the particle-antiparticle reading of the 8-spinor assigns elementary charge $q = \pm \frac{1}{2}$ in units of the gauge coupling g . Consequently the electromagnetic coupling is $e_0 = g/2$ and the fine-structure constant is $\alpha_{\text{em}} = e_0^2/4\pi = g^2/16\pi$, with g the sole free parameter. **[DERIVED]***

Step 1 — the temporal-rotation algebra $\mathfrak{su}(2)_T$

Each L_i squares to -1 , and the three close into $\mathfrak{su}(2)$ with structure constant -2 .

Wolfram Language input

```
Wk = (g[1]+g[2]+g[3])/Sqrt[3];
L1 = g[2].g[3]; L2 = g[3].g[1]; L3 = g[1].g[2];
comm[A_,B_] := A.B - B.A;
Lsq = {L1.L1 === -I8, L2.L2 === -I8, L3.L3 === -I8};
struct = {comm[L1,L2] === -2 L3, comm[L2,L3] === -2 L1, comm[L3,L1] === -2 L2};
{"L_i^2 = -I8" -> Lsq, "structure const -2" -> struct}
```

Literal Wolfram output

```
{"L_i^2 = -I8" -> {True, True, True},
"su(2)_T structure const -2" -> {True, True, True}}
```

Step 2 — no single generator fixes the clock

Because the commutator is insensitive to the positive scalar $1/\sqrt{3}$, it suffices to test $e_1 + e_2 + e_3$. None of L_1, L_2, L_3 commutes with it.

Wolfram Language input

```
clock = g[1]+g[2]+g[3];
singleFix = {comm[L1,clock] === 0, comm[L2,clock] === 0, comm[L3,clock] === 0};
"no single L_i fixes clock ([L_i,clock]=0?)" -> singleFix
```

Literal Wolfram output

```
"no single L_i fixes clock ([L_i,clock]=0?)" -> {False, False, False}
```

Step 3 — the centralizer is one-dimensional

Impose $[aL_1 + bL_2 + cL_3, e_1 + e_2 + e_3] = 0$ and solve the resulting linear system over the matrix entries. The literal `Solve` output is the single family $\{b \rightarrow a, c \rightarrow a\}$: every solution has $a = b = c$, so the centralizer is spanned by $L_1 + L_2 + L_3$.

Wolfram Language input

```
X = a L1 + b L2 + c L3;
eqs = Thread[Flatten[comm[X,clock]] == 0];
cent = Solve[eqs, {a,b,c}];
"centralizer Solve output" -> cent
```

Literal Wolfram output

```
"centralizer Solve output" -> {{b -> a, c -> a}}
```

Step 4 — normalization and the photon J

The democratic sum squares to $-3 \mathbf{1}$, so the unit generator is $J = (L_1 + L_2 + L_3)/\sqrt{3}$, with $J^2 = -\mathbf{1}$, $\text{Tr}(J^2) = -8$, and $[J, \hat{W}] = 0$.

Wolfram Language input

```
Jph = (L1+L2+L3)/Sqrt[3];
{"(L1+L2+L3)^2 = -3 I8" -> ((L1+L2+L3).(L1+L2+L3) === -3 I8),
 "J^2 = -I8" -> (Jph.Jph === -I8),
 "Tr(J^2)" -> Tr[Jph.Jph],
 "[J,Wk] = 0" -> (comm[Jph,Wk] === ConstantArray[0,{8,8}])}
```

Literal Wolfram output

```
{"(L1+L2+L3)^2 = -3 I8" -> True, "J^2 = -I8" -> True,
 "Tr(J^2)" -> -8, "[J,Wk] = 0" -> True}
```

Step 5 — spectrum of J and the Hermitian charge $Q = -iJ$

Since $J^2 = -1$, J acts as a complex structure on the real 8-spinor; its spectrum is $\{(\pm i)^4\}$. The Hermitian charge $Q = -iJ$ has spectrum $\{(\pm 1)^4\}$.

Wolfram Language input

```
specJ = Sort[Eigenvalues[Jph]];
Q      = -I*Jph;
specQ = Sort[Eigenvalues[Q]];
{"spec(J)" -> specJ, "Q=-iJ Hermitian" -> (ConjugateTranspose[Q] == Q),
 "spec(Q)" -> specQ, "Tally spec(Q)" -> Tally[specQ]}
```

Literal Wolfram output

```
{"spec(J)" -> {-I, -I, -I, -I, I, I, I, I}, "Q=-iJ Hermitian" -> True,
 "spec(Q)" -> {-1, -1, -1, -1, 1, 1, 1, 1}, "Tally spec(Q)" -> {{-1, 4}, {1, 4}}}
```

Step 6 — the coupling arithmetic

The unit eigenvalue ± 1 of Q is the charge in units of the coupling *before* the particle–antiparticle identification. Reading the real 8-spinor as a Dirac particle together with its conjugate halves the unit to $q = \pm \frac{1}{2}$, whence $e_0 = g/2$ and $\alpha_{\text{em}} = e_0^2/4\pi = g^2/16\pi$ (note $16\pi = 4 \cdot 4\pi$).

Wolfram Language input

```
e0 = g/2;
alphaEM = Simplify[e0^2/(4 Pi)];
{"alpha_em = e0^2/(4 Pi)" -> alphaEM, "16 Pi = 4*4 Pi" -> (16 Pi == 4*4 Pi)}
```

Literal Wolfram output

```
{"alpha_em = e0^2/(4 Pi)" -> g^2/(16 Pi), "16 Pi = 4*4 Pi" -> True}
```

3 Result triad

(A) Raw Wolfram data anchors

Claim	Quantity	Literal value	Status
$\text{Cl}(3, 3) \cong M_8(\mathbb{R})$	rank of 64 blades	64	[EXACT]
Clifford relation	$\{e_a, e_b\} - 2\eta_{ab}\mathbf{1}$	True	[EXACT]
$\mathfrak{su}(2)_T$ closes	$[L_i, L_j]$ const.	-2	[EXACT]
L_i are complex	L_i^2	-1	[EXACT]
no single fix	$[L_i, \text{clock}] = 0$	$\{F, F, F\}$	[EXACT]
centralizer 1-dim	Solve output	$\{b \rightarrow a, c \rightarrow a\}$	[EXACT]

Claim	Quantity	Literal value	Status
normalization	$(L_1+L_2+L_3)^2$	$-3\mathbf{1}$	[EXACT]
photon complex str.	J^2	$-\mathbf{1}$	[EXACT]
photon Killing norm	$\text{Tr}(J^2)$	-8	[EXACT]
photon fixes clock	$[J, \hat{W}]$	0	[EXACT]
photon spectrum	$\text{spec}(J)$	$\{(\pm i)^4\}$	[EXACT]
charge Hermitian	$Q = -iJ$	True	[EXACT]
charge spectrum	$\text{spec}(Q)$	$\{(\pm 1)^4\}$	[EXACT]
charge unit	q (part./antipart.)	$\pm \frac{1}{2}$	[DERIVED]
coupling	e_0	$g/2$	[DERIVED]
fine structure	α_{em}	$g^2/16\pi$	[DERIVED]
free parameter	g	not fixed	[OPEN]

(B) Mathematical symmetries

The spectrum of Q is symmetric under sign reflection, $\text{spec}(Q) = -\text{spec}(Q)$, with equal multiplicities $4 + 4$; this is the charge-conjugation symmetry realized on the real module. The centralizer condition is invariant under the cyclic S_3 permutation of (e_1, e_2, e_3) , and J (the democratic sum) is the unique fixed direction of that permutation inside $\mathfrak{su}(2)_T$ – the same democratic symmetry that selects the kernel $\hat{W}$. The map $J \mapsto Q = -iJ$ is the real-to-complex structure isomorphism: the imaginary unit of the charge sector is recovered as a bivector, not postulated.

(C) Physical interpretation and boundary limits

The electromagnetic $U(1)$ is not an input: it is the *unique* stabilizer of the physical clock inside the temporal-rotation algebra, fixed by the linear condition of Step 3 and verified as a one-dimensional **Solve** family. Charge quantization $q = \pm \frac{1}{2}$ follows from the fourfold-degenerate ± 1 spectrum of the Hermitian Q together with the particle–antiparticle reading of the eight real components. The chain fixes the *structure* of the charge sector and the relation $\alpha_{\text{em}} = g^2/16\pi$ exactly; it does *not* fix the magnitude of the single coupling g , which remains an irreducible input ([OPEN]). The spectral and algebraic facts ([EXACT]) are representation-independent statements about $\text{Cl}(3, 3)$; the charge *assignment* ([DERIVED]) carries the one physical reading – the particle–antiparticle identification – on which the factor $\frac{1}{2}$ rests.

Summary. The photon generator J is forced uniquely as the centralizer of $\hat{W}$ in $\mathfrak{su}(2)_T$ (literal **Solve** output $\{b \rightarrow a, c \rightarrow a\}$); $\text{spec}(J) = \{(\pm i)^4\}$ and $\text{spec}(-iJ) = \{(\pm 1)^4\}$; the particle–antiparticle reading gives $q = \pm \frac{1}{2}$, hence $e_0 = g/2$ and $\alpha_{\text{em}} = g^2/16\pi$. Everything but the magnitude of g is verified in the 8×8 representation.